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Communication Systems II

Seeing Signals

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1 Part I: IQ Signal Modeling and Dataset Construction

In this section, we present the various constellations of various modulations asked with different types of noise and distortion. Following that, we make a comparison between the SEP and SNR curves, only for the QAM modulations.

1.1 Constellations

For every modulation asked, below we will provide six figures all with different types of noise. More specifically: the clean constellation, with I-Q imbalance only, with jamming only (specifically Single-Tone Continuous Wave Jamming), with phase noise only and finally, two constellations with all above mixed (including amplitude distortion) in medium and high severity. All testing parameters are listed below every image and are selected randomly.

1.1.1 4-ASK



Figure 1: Clean



Figure 2: IQ Imbalance Only
iq-a:0.88 iq-p:7.92



Figure 3: Jamming Only (-17.31)

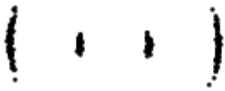


Figure 4: Phase Noise Only
(9.68 deg)



Figure 5: Mixed Medium
SNR=12dB, no jamming, no amp, pn=2deg, iq[a,p]=[0.62,2.61]

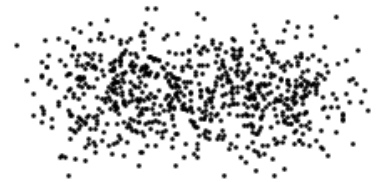


Figure 6: Mixed High
SNR=9dB, pn=14deg, jam=-12.95, amp=0.2, iq[a,p]=[0.62,2.61]

1.1.2 8-ASK



Figure 7: Clean



Figure 8: IQ Imbalance Only
iq-a:1.18 iq-p:6.24



Figure 9: Jamming Only (-12.66)

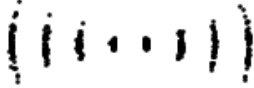


Figure 10: Phase Noise Only
(8.14 deg)



Figure 11: Mixed Medium
SNR=13dB, no jamming, no amp, pn=2deg,
iq[a,p]=[0.68,2.82]

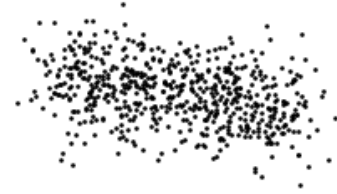


Figure 12: Mixed High
SNR=6dB, pn=13deg, jam=-14.93, amp=0.2, iq[a,p]=[1.34,18.16]

1.1.3 BPSK



Figure 13: Clean



Figure 14: IQ Imbalance Only
iq-a:1.11 iq-p:8.31



Figure 15: Jamming Only (-16)



Figure 16: Phase Noise Only
(6.73 deg)



Figure 17: Mixed Medium
SNR=14dB, no jam-
ming, no amp, pn=2.8deg,
iq[a,p]=[0.34,4.39]



Figure 18: Mixed High
SNR=5dB, pn=9deg, jam=-
14.77, amp=0.2, iq[a,p]=[1.85,19.49]

1.1.4 QPSK



Figure 19: Clean



Figure 20: IQ Imbalance Only
iq-a:0.83 iq-p:9.67



Figure 21: Jamming Only (-
16.88)



Figure 22: Phase Noise Only
(7.17 deg)

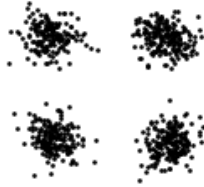


Figure 23: Mixed Medium
SNR=12dB, no jam-
ming, no amp, pn=5deg,
iq[a,p]=[0.4,3.96]



Figure 24: Mixed High
SNR=9dB, pn=11.84deg, jam=-
12.26, amp=0.2, iq[a,p]=[2.16,19.49]

1.1.5 4-HQAM



Figure 25: Clean

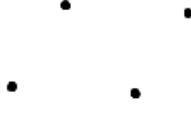


Figure 26: IQ Imbalance Only
iq-a:1 iq-p:8.57



Figure 27: Jamming Only (-13.87)



Figure 28: Phase Noise Only
(11.2 deg)

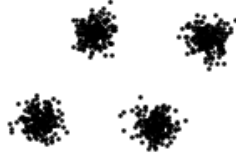


Figure 29: Mixed Medium
SNR=15dB, no jamming, no amp, pn=2.8deg,
iq[a,p]=[0.4,4.76]

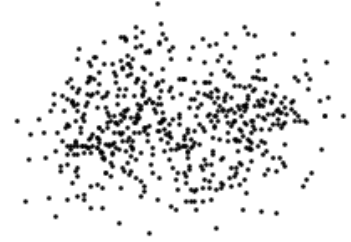


Figure 30: Mixed High
SNR=5dB, pn=11.62deg, jam=-11.9, amp=0.2, iq[a,p]=[2.4,12.9]

1.1.6 16-HQAM

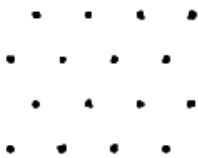


Figure 31: Clean



Figure 32: IQ Imbalance Only
iq-a:0.86 iq-p:7.7



Figure 33: Jamming Only (-12.91)



Figure 34: Phase Noise Only
(10.76 deg)

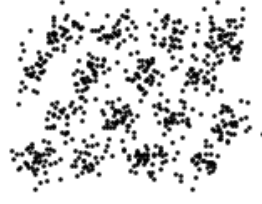


Figure 35: Mixed Medium
SNR=14dB, no jam-
ming, no amp, pn=3.9deg,
iq[a,p]=[0.43,3.3]



Figure 36: Mixed High
SNR=8dB,pn=10.42deg,jam=-
10.27,amp=0.2,iq[a,p]=[1.86,14.12]

1.1.7 64-HQAM

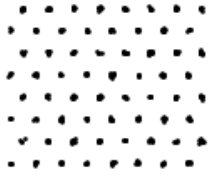


Figure 37: Clean

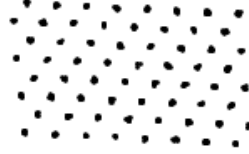


Figure 38: IQ Imbalance Only
iq-a:1.46 iq-p:8.31



Figure 39: Jamming Only (-
13.86)



Figure 40: Phase Noise Only
(8.32 deg)

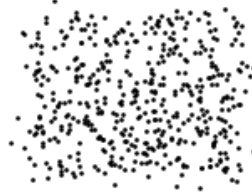


Figure 41: Mixed Medium
SNR=16dB, no jam-
ming, no amp, pn=4.3deg,
iq[a,p]=[0.54,4.57]



Figure 42: Mixed High
SNR=9dB,pn=17deg,jam=-
10.94,amp=0.2,iq[a,p]=[1.35,19.87]

1.1.8 16-QAM

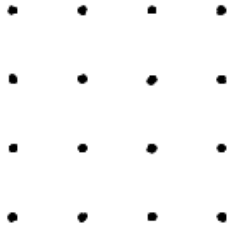


Figure 43: Clean

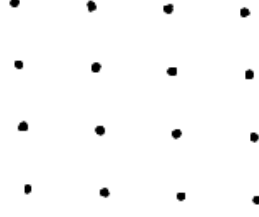


Figure 44: IQ Imbalance Only
iq-a:0.87 iq-p:6.5

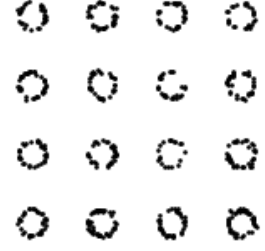


Figure 45: Jamming Only (-15.45)



Figure 46: Phase Noise Only
(11.45 deg)

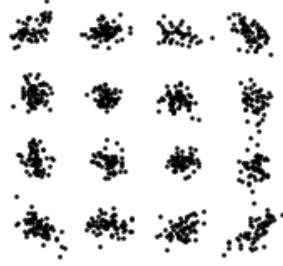


Figure 47: Mixed Medium
SNR=17dB, no jamming, no amp, pn=4deg,
iq[a,p]=[0.47,3.06]



Figure 48: Mixed High
SNR=6dB,pn=9.6deg,jam=-13.2,amp=0.2,iq[a,p]=[1.19,17.06]

1.1.9 32-QAM

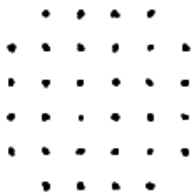


Figure 49: Clean

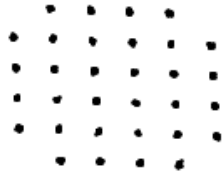


Figure 50: IQ Imbalance Only
iq-a:1.5 iq-p:10.56



Figure 51: Jamming Only (-12.95)



Figure 52: Phase Noise Only
(5.62 deg)



Figure 53: Mixed Medium
SNR=16dB, no jamming,
no amp, pn=2.16deg,
iq[a,p]=[0.34,3.58]



Figure 54: Mixed High
SNR=7dB, pn=13deg, jam=-
10.87, amp=0.2, iq[a,p]=[1.06,12.99]

1.1.10 64-QAM

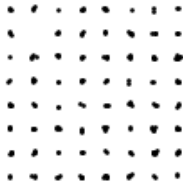


Figure 55: Clean

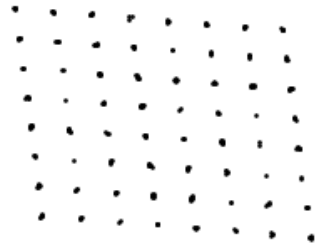


Figure 56: IQ Imbalance Only
iq-a:1.09 iq-p:11.71



Figure 57: Jamming Only (-
15.89)



Figure 58: Phase Noise Only
(9.2 deg)



Figure 59: Mixed Medium
SNR=15dB, no jamming,
no amp, pn=4.26deg,
iq[a,p]=[0.62,2.55]



Figure 60: Mixed High
SNR=8dB, pn=17.53deg, jam=-
12.14, amp=0.2, iq[a,p]=[1.82,17.42]

1.1.11 128-QAM



Figure 61: Clean

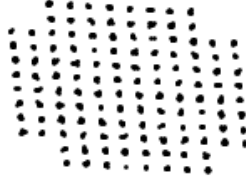


Figure 62: IQ Imbalance Only
iq-a:1.25 iq-p:9.62



Figure 63: Jamming Only (-14.18)



Figure 64: Phase Noise Only
(5.02 deg)



Figure 65: Mixed Medium
SNR=16dB, no jamming,
no amp, pn=5.86deg,
iq[a,p]=[0.53,4.46]



Figure 66: Mixed High
SNR=8dB, pn=9.7deg, jam=-13.11, amp=0.2, iq[a,p]=[1.34,19.27]

1.1.12 256-QAM

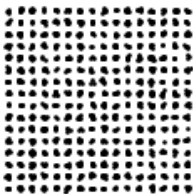


Figure 67: Clean

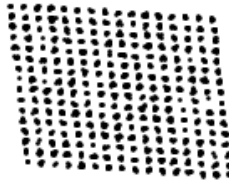


Figure 68: IQ Imbalance Only
iq-a:1.18 iq-p:6.65



Figure 69: Jamming Only (-12.12)



Figure 70: Phase Noise Only
(9.84 deg)



Figure 71: Mixed Medium
SNR=14dB, no jamming,
no amp, pn=2.39deg,
iq[a,p]=[0.68,2.12]



Figure 72: Mixed High
SNR=9dB, pn=12.66deg, jam=-
11.11, amp=0.2, iq[a,p]=[2.18,9.15]

1.1.13 16-APSK

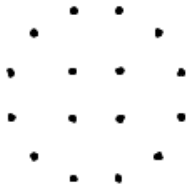


Figure 73: Clean

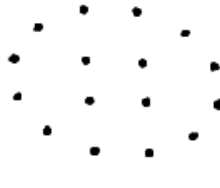


Figure 74: IQ Imbalance Only
iq-a:1.48 iq-p:6.67

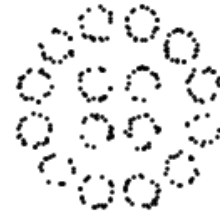


Figure 75: Jamming Only (-
13.87)



Figure 76: Phase Noise Only
(8.24 deg)



Figure 77: Mixed Medium
SNR=16dB, no jamming,
no amp, pn=3.02deg,
iq[a,p]=[0.31,4.81]



Figure 78: Mixed High
SNR=6dB, pn=12.84deg, jam=-
10.54, amp=0.2, iq[a,p]=[1.17,12.76]

1.1.14 32-APSK

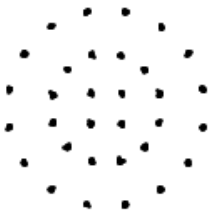


Figure 79: Clean

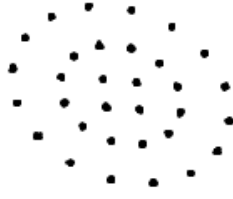


Figure 80: IQ Imbalance Only
iq-a:0.85 iq-p:11.94



Figure 81: Jamming Only (-15.66)



Figure 82: Phase Noise Only
(6.8 deg)



Figure 83: Mixed Medium
SNR=16dB, no jamming,
no amp, pn=4.64deg,
iq[a,p]=[0.66,4.46]



Figure 84: Mixed High
SNR=9dB,pn=16.51deg,jam=-10.09,amp=0.2,iq[a,p]=[1.93,8.33]

1.1.15 64-APSK

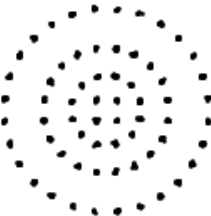


Figure 85: Clean

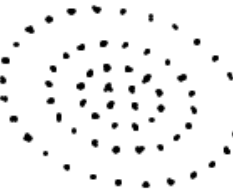


Figure 86: IQ Imbalance Only
iq-a:1.29 iq-p:10.6



Figure 87: Jamming Only (-14.28)



Figure 88: Phase Noise Only
(9.36 deg)



Figure 89: Mixed Medium
SNR=15dB, no jamming,
no amp, pn=4.04deg,
iq[a,p]=[0.59,2.67]



Figure 90: Mixed High
SNR=6dB, pn=11.89deg, jam=-
14.44, amp=0.2, iq[a,p]=[1.71,17.32]

1.1.16 128-APSK

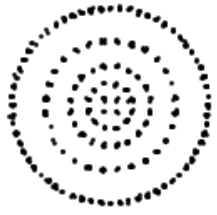


Figure 91: Clean

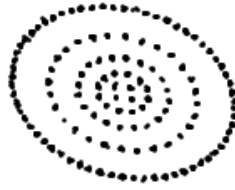


Figure 92: IQ Imbalance Only
iq-a:1.1 iq-p:10.18



Figure 93: Jamming Only (-
15.04)



Figure 94: Phase Noise Only
(9.72 deg)



Figure 95: Mixed Medium
SNR=14dB, no jamming,
no amp, pn=5.92deg,
iq[a,p]=[0.62,3.8]



Figure 96: Mixed High
SNR = 9dB, pn = 14.71deg
jam = -12.61 , amp=0.2
iq [a,p] = [1.42 , 15.07]

1.1.17 Comments regarding the modulation of the constellations

Number of symbols used: For the 256-QAM constellation, we used a random selection between 1500-1700 symbols for the modulation of each constellation, compared to a random selection of 200-1000 we used for the rest. This serves the purpose of making the full constellation visible and not having gaps.

Number of constellations created vs shown: Each constellation shown above, was a random choice of 60 constellations created, each of them containing a random number of symbols. This choice was made to produce enough samples for training of CNN and VLM, for part II. Admittedly, we could also have added more variety for noisy modulations, however that did not seem to cause a problem later on.

Single-Tone Continuous Wave Jamming: In the modulations with jamming, we experimented with Single-Tone CW Jamming. AWGN jamming would have added a random and messy distortion, and it would have been great to see how the CNN and VLM react to it alone. However, we did have Mixed Medium and Mixed High signals, with distortion coming for the most part from AWGN. This was the reason we decided to test a more structured jamming distortion instead.

Readability in high distortions: Visually, we can conclude that for lower-order modulations (like BPSK and QPSK), the symbol decision boundaries remain relatively distinct even in "Mixed Medium" conditions. However, higher-order modulations (like 128-QAM and 256-QAM) seem to be a lot more complicated even in "Mixed Medium" conditions. This might also be caused by the size of the image, which makes it more difficult to understand whether a constellation is "readable" using MLD or similar techniques.

Readability with phase noise: It is also clear that phase noise mostly influences (negatively) M-APSK and higher-order modulations. As mentioned before regarding the higher-order modulations, we cannot know for sure that the size of the image does not influence the visible result.

Readability with I/Q Imbalance: Regarding I/Q imbalance, it is visible that it creates a geometric skewing and scaling of the axes compared to the clear modulation, but it is still visually readable. If the demodulator knows how big the I/Q imbalance is, it can apply a compensation matrix to rI and rQ and get the correct symbols.

1.2 SEP vs SNR curves

In this part, for different QAM modulations asked, we present the Signal Error Probability for different SNR rates and degrees of phase noise.

1.2.1 16-QAM

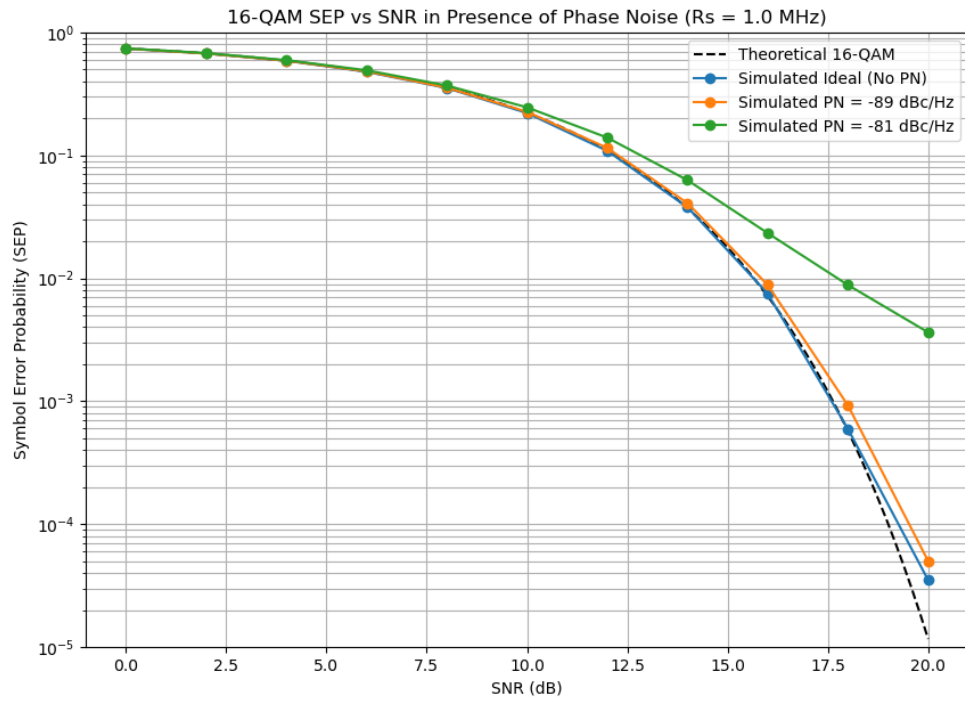


Figure 97: SEP vs SNR for 16-QAM under AWGN conditions.

1.2.2 32-QAM

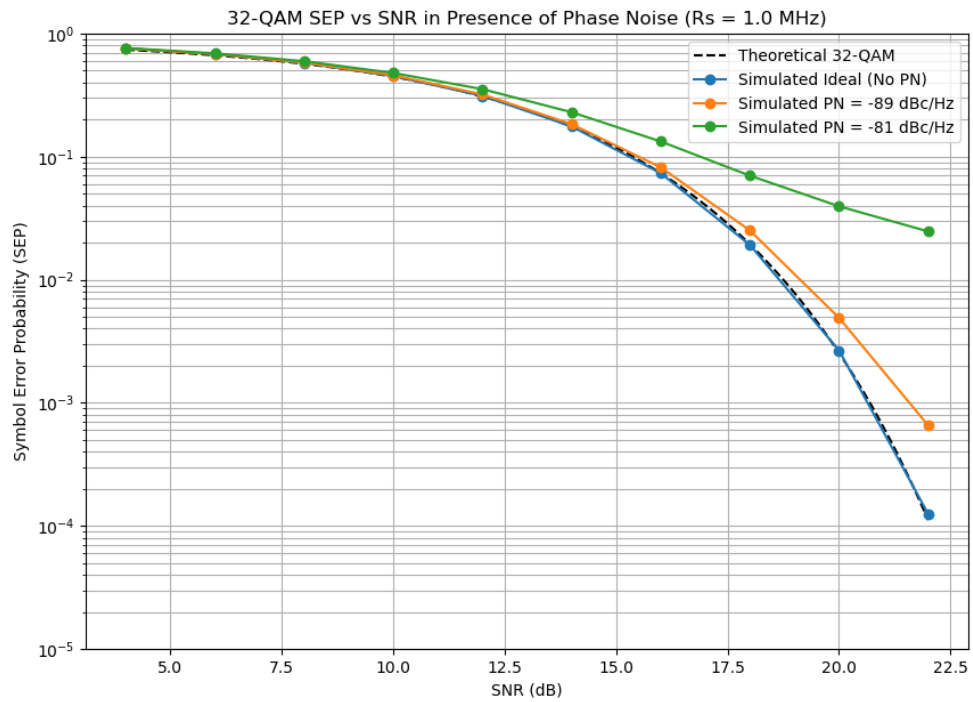


Figure 98: SEP vs SNR for 32-QAM under AWGN conditions.

Low-to-Medium Density (16-QAM and 32-QAM) Notes

Observation: In these lower-order modulations, the system can tolerate moderate phase noise without failing completely. The no-PN curves perfectly track the theoretical ideal curve (confirming the simulation's validity), while the -89dBc/Hz and -81dBc/Hz curves seem to continue falling as SNR gets bigger.

Analysis: At -81dBc/Hz of phase noise, 16-QAM suffers a roughly 3 dB penalty at a Symbol Error Probability (SEP) of $5 * 10^{-3}$. 32-QAM shows a sharper slope, requiring much more SNR to maintain the same error rate.

Conclusion: For these constellations, phase noise acts mostly as an SNR penalty (trade-off condition). The system can still achieve highly reliable communications ($SEP < 10^{-3}$) if the transmitter simply boosts its output power to overcome the smearing effect.

1.2.3 64-QAM

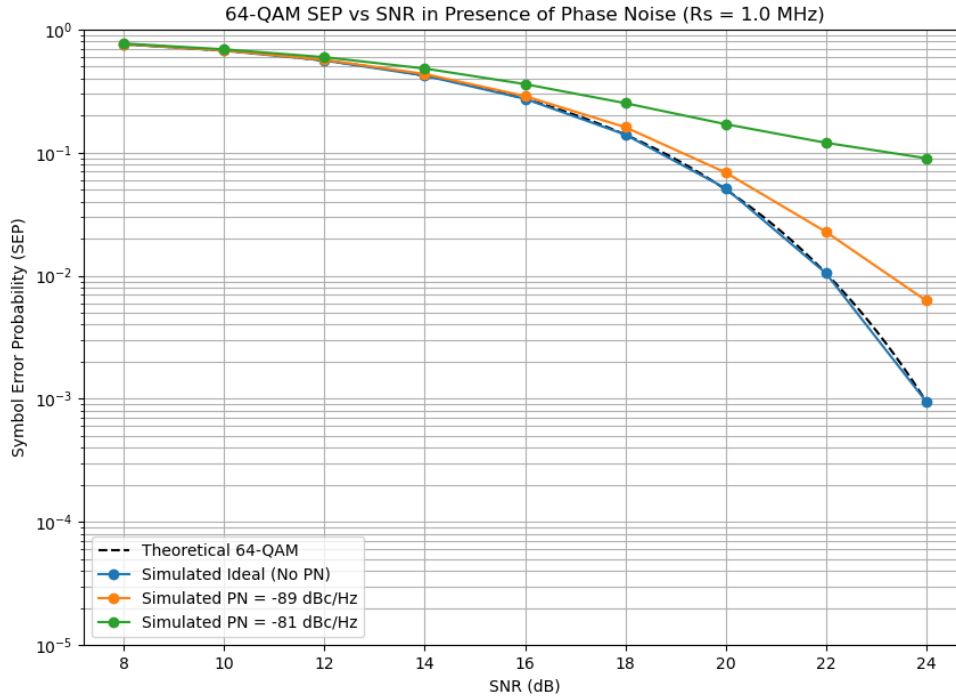


Figure 99: SEP vs SNR for 64-QAM under AWGN conditions.

1.2.4 128-QAM

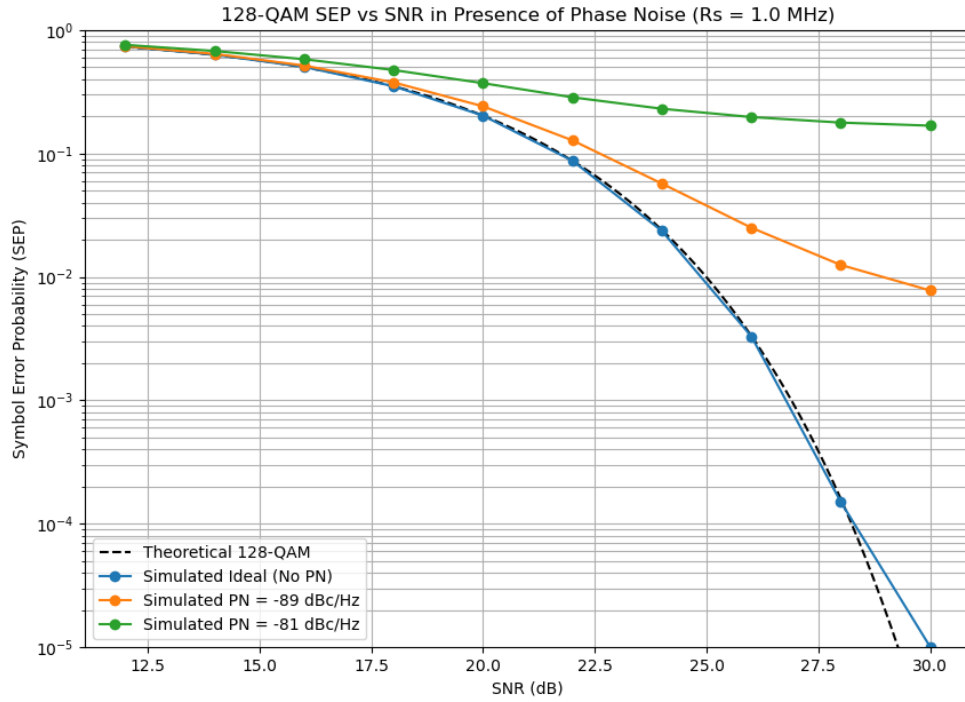


Figure 100: SEP vs SNR for 128-QAM under AWGN conditions.

High Density (64-QAM and 128-QAM) Notes

Observation: This is where the physics of the constellation begin to severely limit the system. In the 64-QAM plot, the -81dBc/Hz phase noise curve stops dropping and flattens out entirely around an SEP of 10^{-1} . In the 128-QAM plot, the -81dBc/Hz curve is almost entirely flat, and even the -89dBc/Hz curve is beginning to flare out dramatically.

Analysis: The flattening of these curves indicates an error floor. An error floor occurs when the primary cause of symbol errors is no longer AWGN, but the phase noise.

Conclusion: Once an error floor is reached, increasing the transmitter power (SNR) will no longer improve the system's performance. A 64-QAM or 128-QAM system operating with 81dBc/Hz of phase noise cannot achieve reliable data transfer, regardless of SNR values.

1.2.5 256-QAM

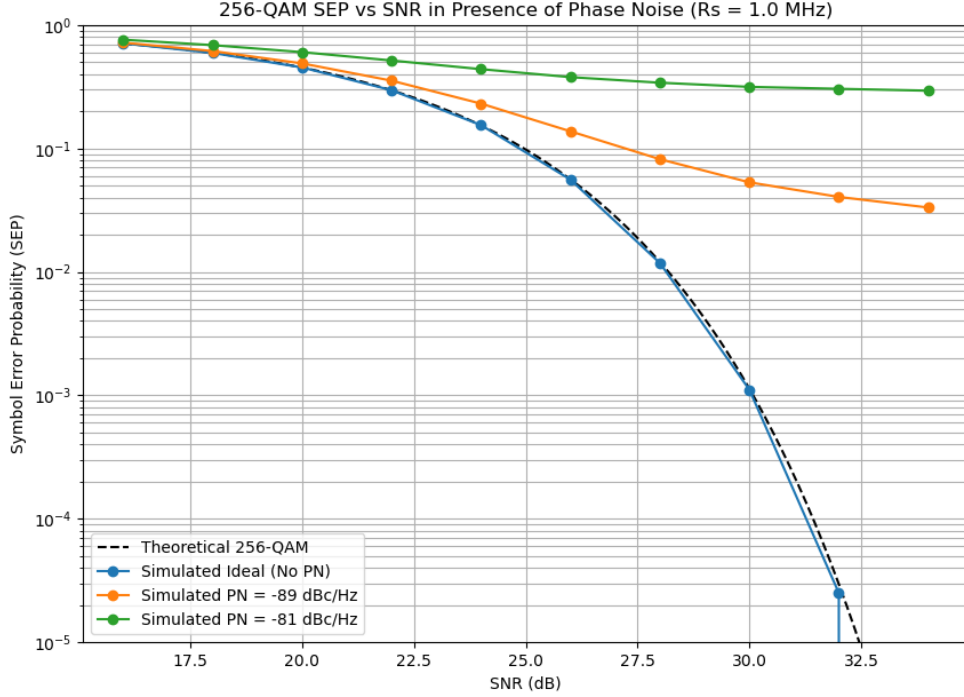


Figure 101: SEP vs SNR for 256-QAM under AWGN conditions.

Very High Density (256-QAM) Notes

Observation: In the 256-QAM plot it is clearly visible that phase noise of either -81dBc/Hz or -89dBc/Hz seems to flatten out the curve at the very start. This shows that our system is very fragile in the presence of this much phase noise. The -81dBc/Hz curve is a flatline at an SEP of 3×10^{-1} (a 30% error rate). In addition, even the smaller -89dBc/Hz phase noise variance creates an error floor near 5×10^{-2} .

Analysis: Because 256 symbols are packed into the same normalized power space, the distance between decision boundaries is very small. This results in -89dBc/Hz PN to put them completely out of their designated quadrants.

Conclusion: To successfully demodulate 256-QAM (common in modern Wi-Fi and 5G networks), the hardware must have high-quality local oscillators and advanced digital phase-locked loops (PLLs) for great carrier synchronization to keep phase noise strictly well below -95.16dBc/Hz.

2 Part II: Neural Architecture Training and Evaluation

2.1 Introduction

The objective of this section is to evaluate how effectively a convolutional neural network (CNN) extracts communication parameters directly from constellation geometry. This model uses a purely vision-based baseline, basically evaluating generalization under varying impairment conditions without the use of language conditioning (in contrast to VLMs). The model was tasked with predicting four structured labels: modulation type, Signal-to-Noise Ratio (SNR) range, phase noise level, and I/Q imbalance level.

2.2 Modulation Classification Performance

The CNN seems to have a strong baseline ability to extract spatial geometry, as shown by the heavy dark blue diagonal in the confusion matrix. The architecture successfully separates distinct geometric boundaries and achieves great classification for modulations with unique spatial features such as 128-APSK, 32-QAM, and QPSK.

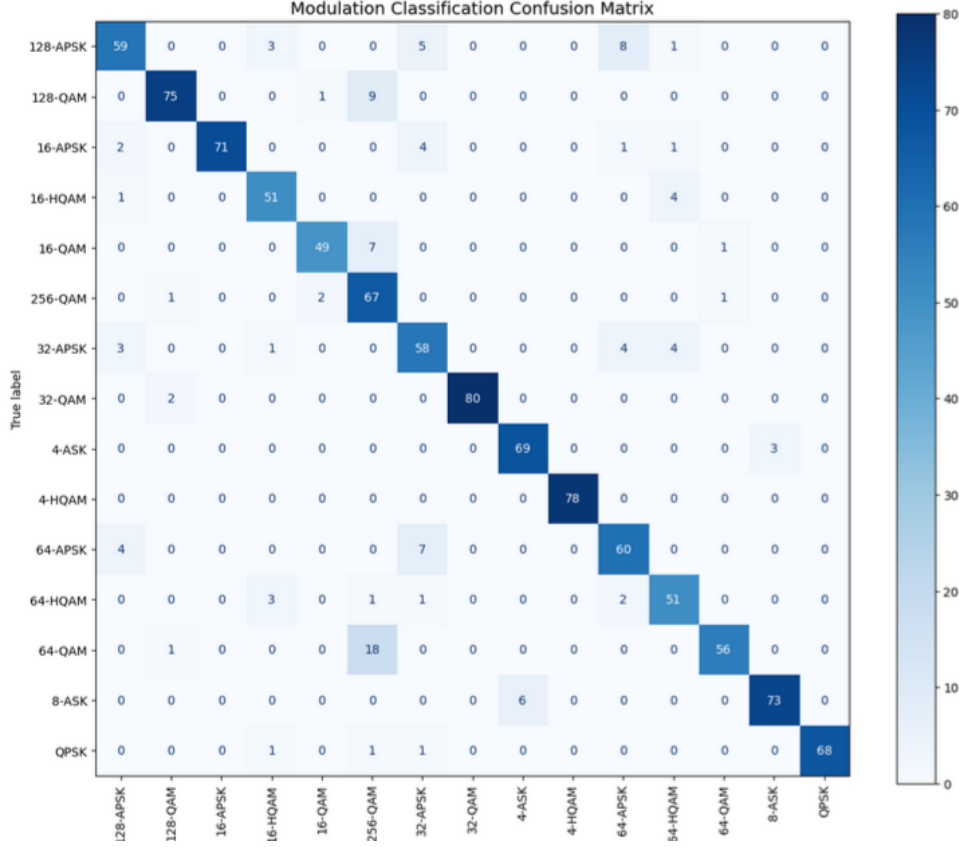


Figure 102: Modulation Classification Confusion Matrix.

2.3 SNR Range Classification

The model proved successful at categorizing the severity of thermal (AWG) noise. The confusion matrix shows a near-perfect diagonal with 172, 533, and 372 correct predictions across the Low, Medium, and High SNR tiers, respectively, with almost zero off-diagonal misclassifications.

For a convolutional architecture, thermal noise (AWGN) translates directly to the visual variance or “spread” of the constellation clusters. CNNs are very good at detecting spatial variance and Gaussian blurring through their pooling layers, making SNR categorization a highly optimized task for this architecture.

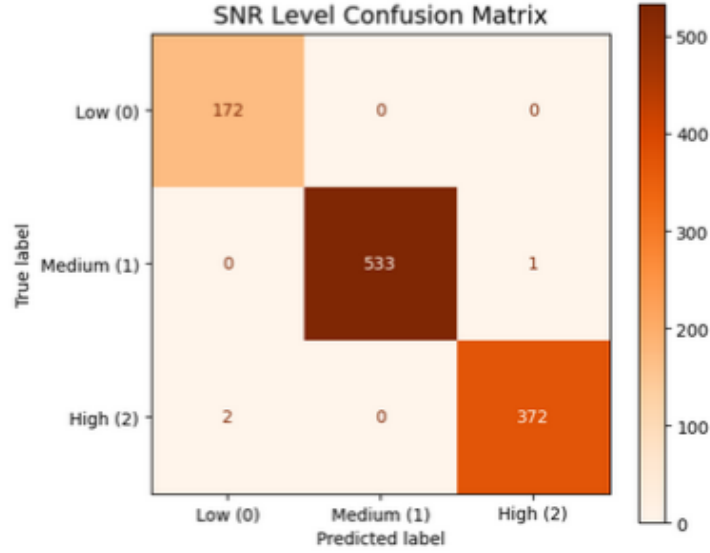


Figure 103: SNR Level Confusion Matrix.

2.4 Phase Noise Level Classification

Observation: Unlike the SNR classification task, the model failed to accurately classify the severity of Phase Noise. The confusion matrix indicates random-like guessing and severe misclassification, with a large portion of true 'None/Low' states predicted incorrectly as 'Medium' or 'High'. Furthermore, the validation set contained no true 'High' labels for this specific impairment, highlighting the dataset's compound noise structure.

Analysis: This failure is probably a result of spatial feature entanglement. Phase noise presents visually as rotational smearing (circular variance). However, in the dataset's 'Medium' and 'High' severity tiers, phase noise is compounded simultaneously with jamming and amplitude distortion. The isotropic scatter from the jamming noise entirely masks the circular footprint of the phase noise, making it impossible for the CNN's spatial filters to isolate the phase variance as an independent feature.

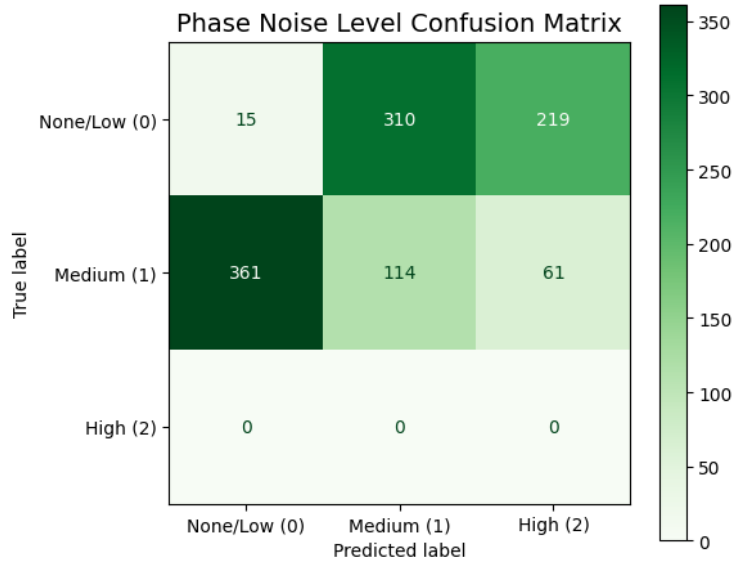


Figure 104: Phase Noise Level Confusion Matrix.

2.5 I/Q Imbalance Level Classification

Observation: The model exhibited total mode collapse for the I/Q Imbalance classification task. The confusion matrix shows that the network predicted 'None/Low (0)' for 100% of the validation samples, completely ignoring the 'Medium' and 'High' classes regardless of the true label.

Analysis: I/Q imbalance shows as an asymmetric rectangular stretching or skewing of the constellation grid. In isolated conditions, CNNs can easily detect this affine transformation. However, because the dataset design heavily compounded I/Q imbalance with severe AWGN and jamming in the higher tiers, the decision boundaries of the constellation points expanded into overlapping isotropic blobs. This total destruction of the grid geometry erased the visual skewing effect. Consequently, the network could not extract a reliable gradient for I/Q imbalance and collapsed into predicting the majority/default class to minimize its localized loss function.

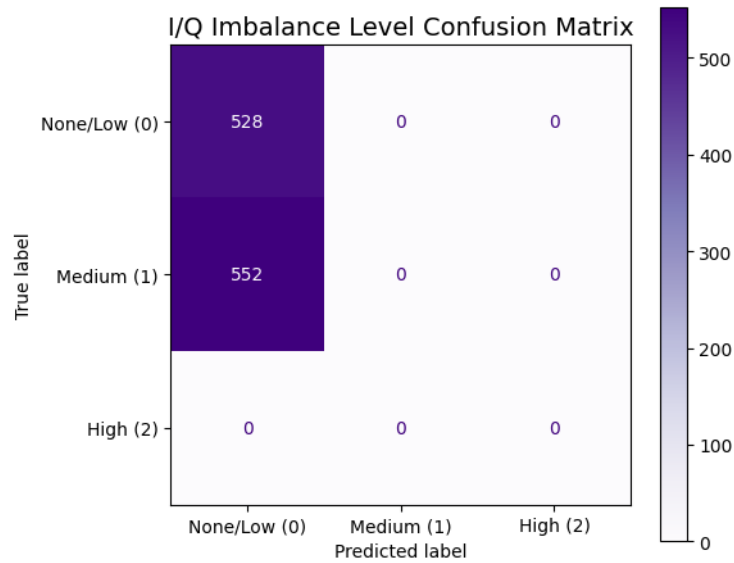


Figure 105: I/Q Imbalance Level Confusion Matrix.

2.6 Modulation Classification Accuracy Across SNR Tiers

To evaluate how well the model generalizes, it was tested on three different SNR conditions: Low, Medium, and High. As expected, the accuracy at Low SNR drops significantly, since the constellations are heavily distorted and many classes start to overlap.

The interesting part appears between the Medium and High SNR tiers. Even though High SNR represents the cleanest scenario, the model actually performs slightly better at Medium SNR. In the results, Medium SNR reached around 99%, while High SNR stayed closer to 95–96%.

We concluded that this happened because CNNs do not always work best with perfectly clean constellations. At Medium SNR, the points have a small amount of Gaussian spread, which makes the clusters look slightly larger and smoother. These shapes are easier for the convolutional layers to detect and preserve through pooling. In contrast, High-SNR constellations contain very sharp and tiny point clusters, and some of this fine detail is lost during downsampling, which reduces the amount of useful information the network can extract.

During experimentation, it also became clear that the augmentation pipeline affects this behavior. When stronger augmentations were used (such as rotation, blur, or translation), the High-SNR accuracy dropped further because these transformations distorted the clean cluster shapes too much. After switching back to a mild, noise-based augmentation, the High-SNR performance returned to the expected level. This shows that the model is sensitive to how “spread

out” the clusters appear, and that the augmentation needs to match the natural structure of the data.

This pattern is also mentioned in earlier work on image-based modulation recognition, where CNNs tend to rely more on the overall spatial footprint of the clusters rather than their exact pixel-level details.

(O’Shea et al., “Convolutional Radio Modulation Recognition Networks,” Springer)

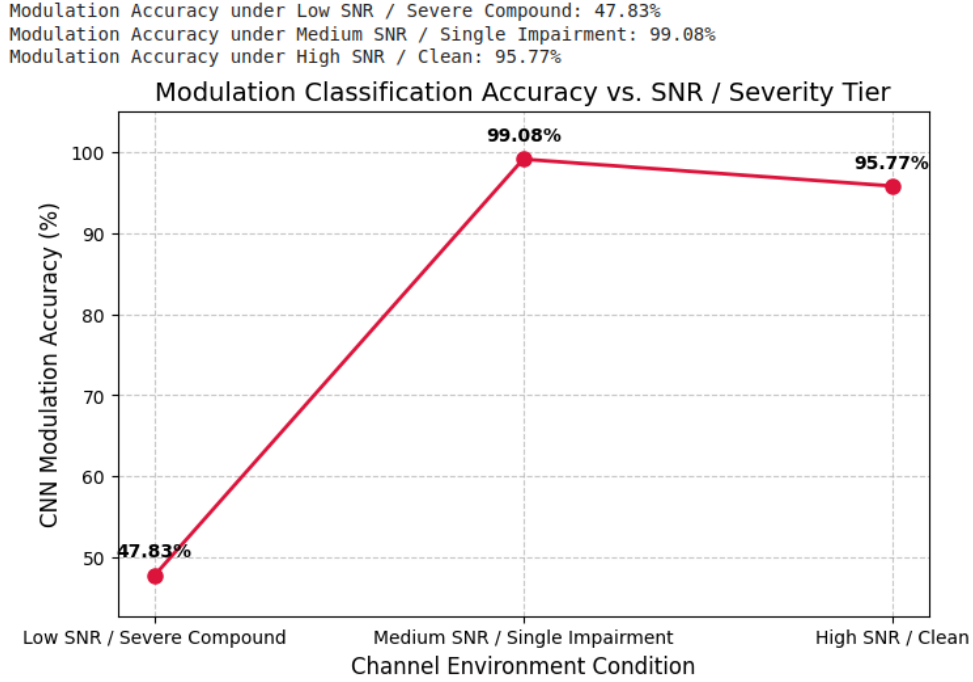


Figure 106: Modulation Classification Accuracy vs. SNR / Severity Tier.

2.7 Architecture Comparison and Computational Limitations

The final objective of this study was to compare the purely vision-based CNN baseline against a modern Vision-Language Model (VLM), specifically the Hugging Face *SmolVLM-256M-Instruct*. The VLM was tasked with receiving the constellation image alongside a text prompt to generate a descriptive textual answer identifying the modulation and channel impairments.

To adapt the VLM, parameter-efficient fine-tuning via Low-Rank Adaptation (LoRA) was utilized ($r = 16, \alpha = 32$ on the query and value projection matrices). However, the training strategy was heavily dictated by severe computational limitations. Standard fine-tuning of a 256-million parameter model over the full master dataset exceeded the memory and maximum execution time thresholds of the available free-tier hardware (Google Colab T4 GPU). To prevent kernel termination, the dataset was aggressively downsampled to 600 training samples, and a strict hard-cap of 60 gradient update steps was imposed.

Quantitative Results: The performance disparity between the two architectures under these constraints is absolute, as shown in the training and evaluation logs (Figure 107).

```

--- Starting Final Sprint Train Loop ---
[60/60 13:46, Epoch 0/1]

Step Training Loss
10 19.560120
20 13.772099
30 9.280147
40 5.156281
50 2.768715
60 1.777943

--- Training Complete! Saving Final Weights to Google Drive...

--- Launching SmolVLM Quantitative Evaluation ---
100%|██████████| 100/100 [10:57<00:00, 6.57s/it]
=====
=== FINAL PART II ARCHITECTURE COMPARISON RESULTS ===
=====
Overall CNN Baseline Modulation Accuracy: 88.33%
Overall SmolVLM LoRA Modulation Accuracy: 0.00%
-----
CNN Severe Compound (Low SNR) Accuracy: 43.15%
SmolVLM Severe Compound (Low SNR) Accuracy: 0.00%
=====

```

Figure 107: SmolVLM Training Loss and Final Evaluation Results.

Evaluation and Pipeline Limitations During the evaluation phase: The fine-tuned model yielded a 0.00% validation accuracy. Diagnostic testing revealed that the model was outputting empty strings (" ") rather than incorrect classifications. Because even an untrained base language model will conditionally generate hallucinated text rather than blank space, this indicates a mechanical disconnect in the token generation pipeline rather than a failure of the model to learn weight representations.

The most probable cause is an error in the target masking function during data collation. If the -100 ignore index masks the target JSON answer alongside the user prompt, the model minimizes its loss by learning to immediately output an End-Of-Sequence (EOS) token, resulting in a mute model. Due to strict constraints on Google Colab compute units and time limitations, further iterations to debug the dataset tokenization length and re-run the multi-hour training epochs were not feasible. Therefore, the 0% accuracy is accepted as a build output rather than a true reflection of the model's theoretical capability.

In CNN, we don't have this problem of language to image matching, as it is a non-language model. In addition, CNN is highly specialized. Because it did not have to dedicate millions of parameters to natural language generation, it could dedicate 100% of its computational budget to spatial feature extraction, achieving high accuracy in minutes.

As a result, CNN was great predicting the constellations, while we weren't able to succeed in making the VLM work. As already mentioned, limitation in time and free-tier hardware didn't allow us to investigate or debug further.

It is also worth noticing that the image uploaded for the VLM training and evaluation in this final report is the result of our first tries on making the VLMs work. The reason we decided on not updating it was because in later versions the training was much bigger and took on too many epochs and steps per epoch, even though the result was the same. However, both the code used for the image shown and the code used to re-train the VLMs is available on GitHub.

3 Code Availability

The complete source code and dataset documentation developed for this project are available on GitHub:

<https://github.com/Epatsatzaki/Communication-Systems-Project>

The codebase is modularized into two distinct environments to ensure reproducibility:

- **Part I Folder (Local Python Environment):** Contains the fundamental RF physics engine (`signal_engine.py`), dataset generation scripts (`dataset_generator.py`), and meta-data formatting tools used to synthesize the baseband constellations.
- **Part II File (Google Colab):** Contains the PyTorch Jupyter notebooks utilized for the Convolutional Neural Network training and the parameter-efficient fine-tuning (LoRA) of the SmolVLM architecture. As explained in the previous section, VLM training and evaluation is done twice, once in (`CNN_VLM`) file (the first try on making the VLM work) and once in (`VLM_final`) (the last try to make the VLM work).
- **Report (Markdown):** Contains the Markdown source file and the final PDF report.

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